EDUAI ACADEMIC SMART NOTES

Generated: October 29, 2026

University Curriculum Series

Introduction to Machine Learning

Section Topic: Overfitting, Underfitting & Advanced Regularization

Executive Summary: Exhaustive treatment of the bias-variance tradeoff, mathematical derivation of L1 Lasso and L2 Ridge regularizations, and concrete anti-overfitting strategies.

1. Core Learning Objectives

- Deconstruct Generalization Error into Bias, Variance, and Irreducible Noise
- Formulate L1 Lasso (sparsity) and L2 Ridge (weight decay) penalties
- Understand ElasticNet compromise and modern deep learning regularization (Dropout, Early Stopping)
- Apply actionable remediation strategies when overfitting or underfitting occurs in production

2. Conceptual Breakdown & Technical Formulations

1. Bias-Variance Decomposition

For any target y = f(x) + \epsilon with irreducible error \epsilon \sim \mathcal{N}(0, \sigma^2), the expected Mean Squared Error of estimator \hat{f}(x) decomposes mathematically:

\mathbb{E}\left[(y - \hat{f}(x))^2 \right] = \text{Bias}[\hat{f}(x)]^2 + \text{Var}[\hat{f}(x)] + \sigma^2

- **Bias**: Error introduced by approximating a complex physical reality with a simplified algorithm. (Underfitting)
- **Variance**: Error stemming from sensitivity to small fluctuations in the training dataset. (Overfitting)
- **\sigma^2**: Inherent irreducible noise within the problem formulation or measurements.

2. Regularization Formulations

To combat high variance, we penalize large parameter magnitudes by augmenting the empirical loss function:

\mathcal{L}_{reg}(\theta) = \mathcal{L}_{data}(\theta) + \lambda \cdot \Omega(\theta)

Where \lambda \ge 0 balances empirical fit against model simplicity.

Ridge Regression (L2 Penalty)

\Omega_{L2}(\theta) = \frac{1}{2} \sum_{j=1}^d \theta_j^2 = \frac{1}{2} \| \theta \|_2^2

- Shrinks weights smoothly towards zero.
- Admits a closed-form analytical solution: \theta_{Ridge} = (X^T X + \lambda I)^{-1} X^T y.
- Retains all features; mitigates multicollinearity.

Lasso Regression (L1 Penalty)

\Omega_{L1}(\theta) = \sum_{j=1}^d |\theta_j| = \| \theta \|_1

- Possesses sharp diamond-shaped contour constraints at axes intersections.
- Forces non-informative coefficients strictly to **zero**, acting as automatic feature selection.

ElasticNet

Combines both penalties via a mixing ratio \rho:

\Omega_{EN}(\theta) = \lambda \left[\rho \| \theta \|_1 + \frac{1 - \rho}{2} \| \theta \|_2^2 \right]

3. Comparison Matrix

Technique	Penalty Formula	Effect on Weights	Feature Selection?	Best Use Case
Ridge (L2)	\sum \theta_j^2	Shrinks uniformly	No	Multicollinear features
Lasso (L1)	\sum |\theta_j|	Drives weights to 0	**Yes (Sparse)**	High-dimensional sparse data
ElasticNet	\alpha L1 + (1-\alpha) L2	Group sparsity	**Yes**	Correlated feature groups

4. Modern Regularization in Deep Learning

- **Dropout**: Inverted dropout randomly zeroes neuron activations with probability p during forward propagation, preventing complex co-adaptations.
- **Early Stopping**: Halts gradient updates when validation error ceases to improve for k consecutive epochs.
- **Data Augmentation**: Artificially inflates dataset size via domain-specific transformations (flips, rotations, jitter).

5. Practical Python Demonstration

```
from sklearn.linear_model import Ridge, Lasso, ElasticNet
ridge = Ridge(alpha=1.0)
lasso = Lasso(alpha=0.1)
elastic = ElasticNet(alpha=0.5, l1_ratio=0.5)
ridge.fit(X_train, y_train)
lasso.fit(X_train, y_train)
elastic.fit(X_train, y_train)
print(f"Lasso Zeroed-out Features: {np.sum(lasso.coef_ == 0)} of {X_train.shape[1]}")
```

6. Exam-Important Concepts & Review Checklist

- Why does Lasso produce exact zeros while Ridge does not? The L1 norm geometry has sharp corners that intersect the error elliptical contours on the coordinate axes.
- Increasing \lambda increases Bias and decreases Variance.
- Decreasing \lambda decreases Bias and increases Variance.

3. Self-Assessment & Exam Checklist

Topic Milestone	Mastery Check
Can you formulate the primary mathematical loss / model objective?	[ ] Confirmed

Can you distinguish parametric vs non-parametric assumptions?	[ ] Confirmed
Are you prepared to calculate precision, recall, and F1 metrics?	[ ] Confirmed
Do you understand the difference between L1 sparsity and L2 shrinkage?	[ ] Confirmed